

Introduction to genetic simulation for animal breeding and genetics

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Agenda

1. Introduction to the course and practical matters
2. Motivation for simulations
3. Approaches to genetic simulation
4. Demo in AlphaSimR

Practical matters

P000172, Simulation for Animal Breeding: Genetics, Biomarkers, and Disease transmission

1.5 Hp

Objectives

On completion of the course, the student should be able to:

- Design and implement breeding strategy models
- Use relevant source code for data simulation
- Critically assess simulation settings and parameters
- Analyse and interpret simulation outcomes

Schedule

Day 1 🌞 (21 Sep)

09:15–12:00: Introduction lecture - Infinite applications of simulations (MJ)

13:00–16:30: Simulating phenotypic data   (PDC)

Activities: **lectures** and **computer exercises**

All **materials** appear on the course website

Plus **recommended literature** for self studies

Day 2 🌞 (22 Sep)

Simulating breeding programs 🐄 🐷 (PDC & HM)

Day 3 🌞 (23 Sep)

Simulating breeding programs 🐔 🐟 (MJ)

Day 4 🌞 (24 Sep)

Simulating disease transmission 🦠 (HM)

Day 5 🌞 (25 Sep)

Simulating biomarker and RNA-Seq data 🧬 (HM)

Power analysis and statistical power testing (MJ)

Day 6 🍁 (1 Oct; 💻)

Individual project presentations (1/2)

Day 7 🍁 (2 Oct; 💻)

Individual project presentations (2/2)

Examination: individual project

- Build a simulation that tackles some genetic problem of interest to you.
- It may relate to any of the topics of the course (breeding programs, disease, genomic data, biomarkers, power analysis).
- **Hand in code** and short **documentation** (specific instructions later)
- **Oral presentation** in Zoom seminar 1 or 2 October (schedule later), also participate in part of the seminar

Think about a project that you can do in **2 days**.

Getting to know you (Menti)

Motivation for simulations

Modelling

Computational modelling

Simulation

(Scientific) modelling: representing a system and its behaviour in a simplified, but hopefully useful, way.

Intermediate between theory and data — models are instruments that help us relate theory and data (Morrisson & Morgan 1999)

Computational modelling: mathematical modelling that involves computation.

(Stochastic) simulation: computational model with element of randomness

Suggested reading: "Introduction to genetic simulation", course homepage.

The pizza problem

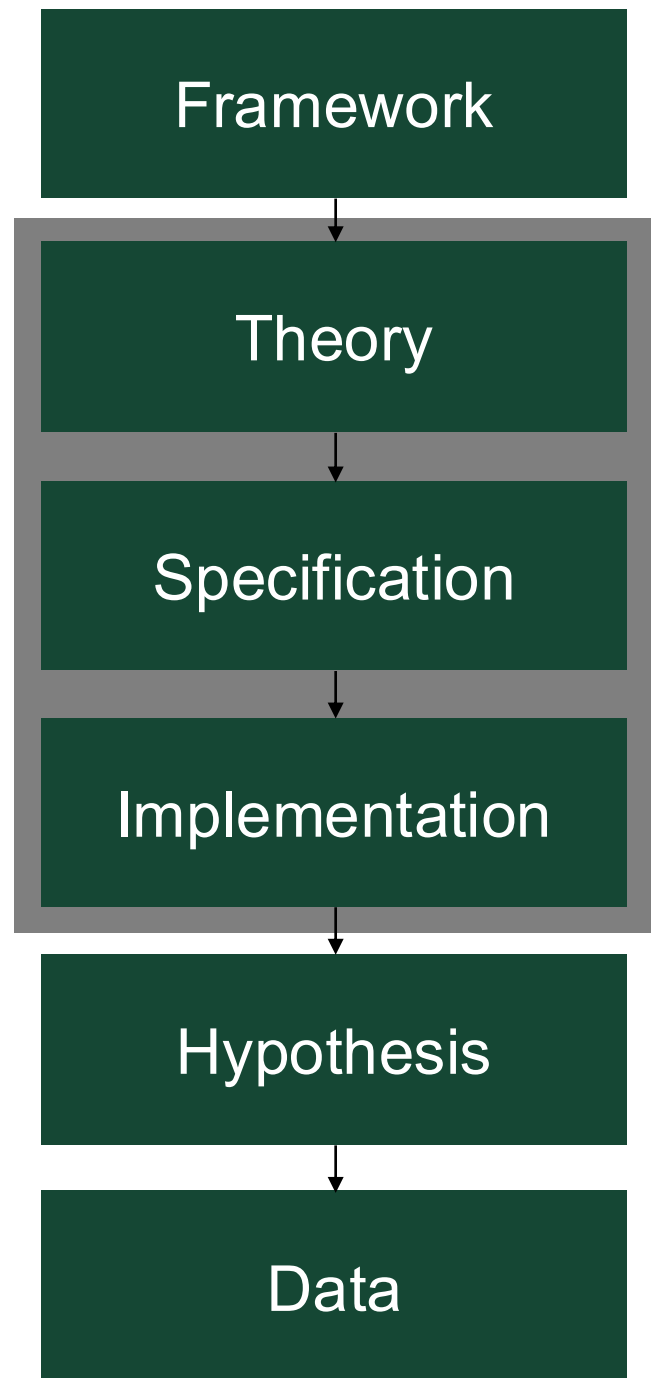
'A Twitter post ([Fermat's Library, 2019](#)) said "a useful counterintuitive fact: one 18 inch pizza has more 'pizza' than two 12 inch pizzas"—along with an image similar to [Figure 1](#). The reaction to this tweet was largely surprise or disbelief. For example, one follower replied, "But two pizzas are more than one" ([Sykes, 2019](#)).'



(Guest & Martin 2021)

Toy example: how do we model this?

“To start, one must create (a) a verbal description, a conceptual analysis, and/or a theory; (b) a formal (or formalizable) description, that is, a specification using mathematics, pseudocode, flowcharts, and so on; and (c) an executable implementation written in programming code”



Modelling process

”most often ignored or left out from research descriptions”

(Guest & Martin 2021)

Assume we want the greatest volume of food.

Mathematical model for volume of pizza order of N pizzas with radius r_j

$$\psi_i = \sum_{j=1}^N \pi r_j^2$$



(Guest & Martin 2021)

$$\psi_i = \sum_{j=1}^N \pi r_j^2$$

```
pizza_order <- function(pizza_diameters) {  
  sum(pi * (pizza_diameters/2)^2)  
}
```

```
pizza_order(c(12, 12))  
## [1] 226.1947
```

```
pizza_order(18)  
## [1] 254.469
```

(Guest & Martin 2021)

It is hard to intuitively draw out the implications of theories in our heads.

Even when the cases are super simple.

Even when there is very established mathematical theories.

We need to do the calculations.

Simulation is widely used in animal breeding and genetics.

Here are a few examples from the literature on **genomic selection**, some **method development**, and **breeding strategy** work.

Prediction of Total Genetic Value Using Genome-Wide Dense Marker Maps

T. H. E. Meuwissen,^{*} B. J. Hayes[†] and M. E. Goddard^{†,‡}

”Recent advances in molecular genetic techniques will make dense marker maps available and genotyping many individuals for these markers feasible. Here we attempted to estimate the effects of ~50,000 marker haplotypes simultaneously from a limited number of phenotypic records. A genome of 1000 cM was simulated with a marker spacing of 1 cM. The markers surrounding every 1-cM region were combined into marker haplotypes. Due to finite population size ($N_e = 100$), the marker haplotypes were in linkage disequilibrium with the QTL located between the markers. ... It was concluded that selection on genetic values predicted from markers could substantially increase the rate of genetic gain in animals and plants, especially if combined with reproductive techniques to shorten the generation interval.”

(Meuwissen, Hayes & Goddard 2001)

There were several early “genomic selection papers”. Another early simulation:

Nejati-Javaremi A, Smith C, Gibson J. Effect of total allelic relationship on accuracy of evaluation and response to selection. Journal of animal science. 1997;75: 1738–1745.
doi:10.2527/1997.7571738x

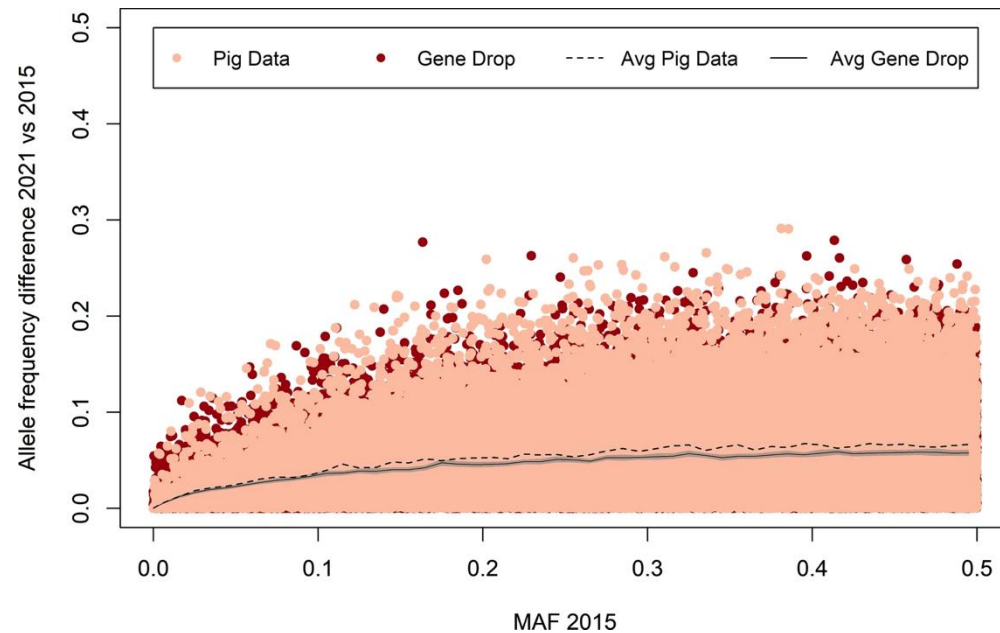
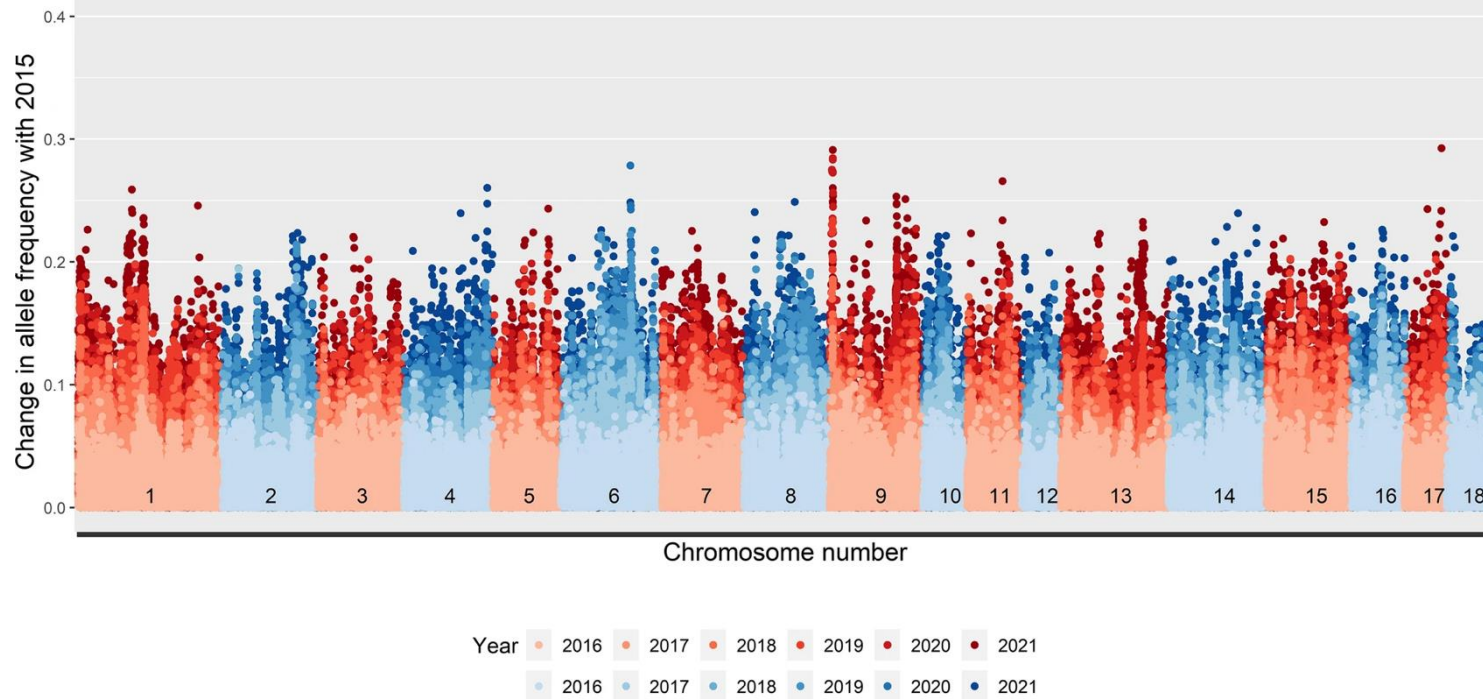
Is genomic selection accuracy driven by **old linkage disequilibrium** in founders, or by close **relationships within the breeding population**?

Habier D, Fernando RL, Dekkers JCM. The Impact of Genetic Relationship Information on Genome-Assisted Breeding Values. Genetics. 2007;177: 2389–2397. doi:10.1534/genetics.107.081190

Artificial simulation scenarios with GBLUP with LD and no LD.

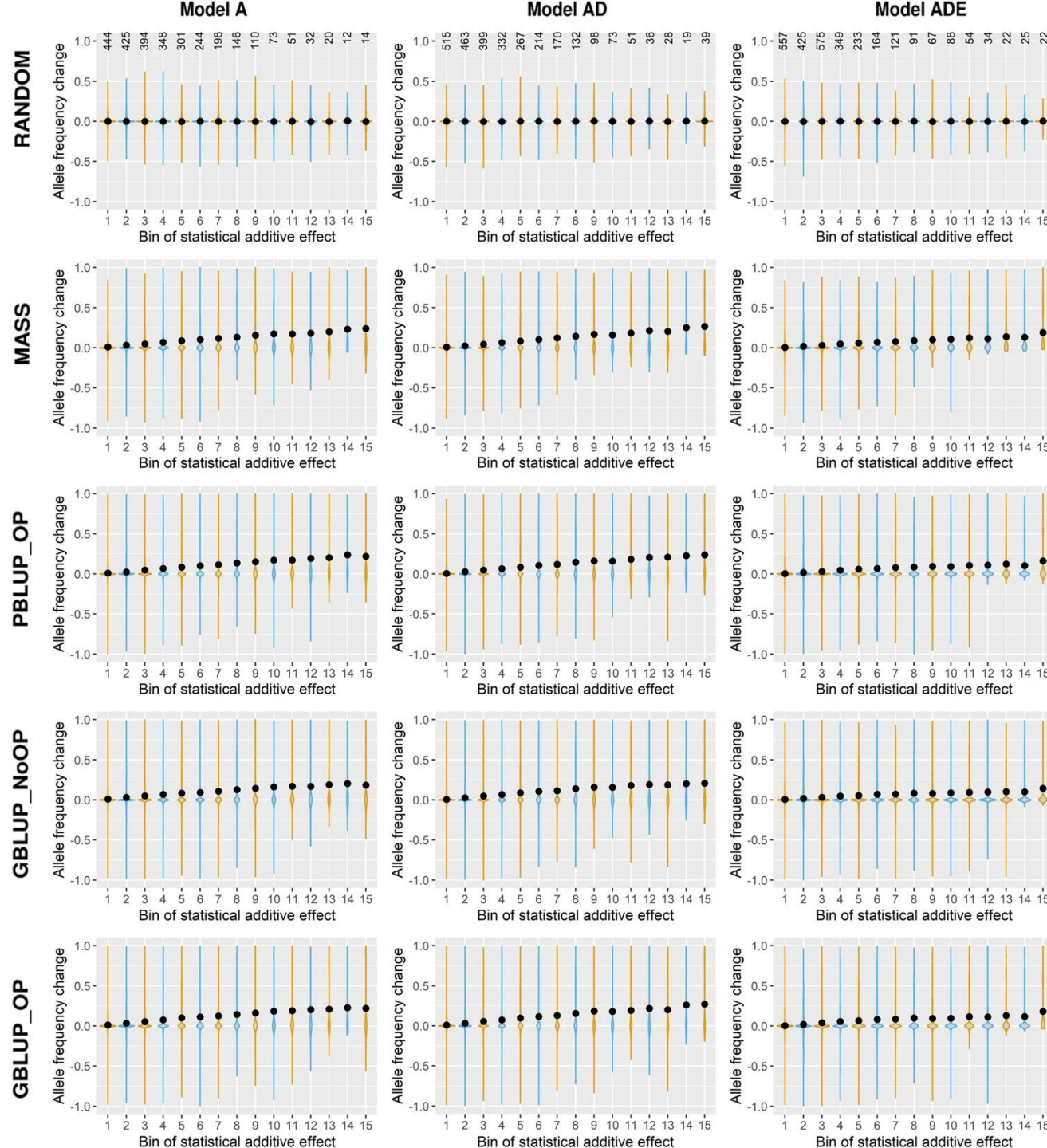
”To show that the accuracy of GEBVs is nonzero even if there is no LD in the population, 100, 1000, and 2000 markers in linkage equilibrium (LE) with 10 QTL were simulated. To ensure linkage equilibrium between markers and QTL, each locus (markers and QTL) was located on a different chromosome.”

(Habier et al. 2007)

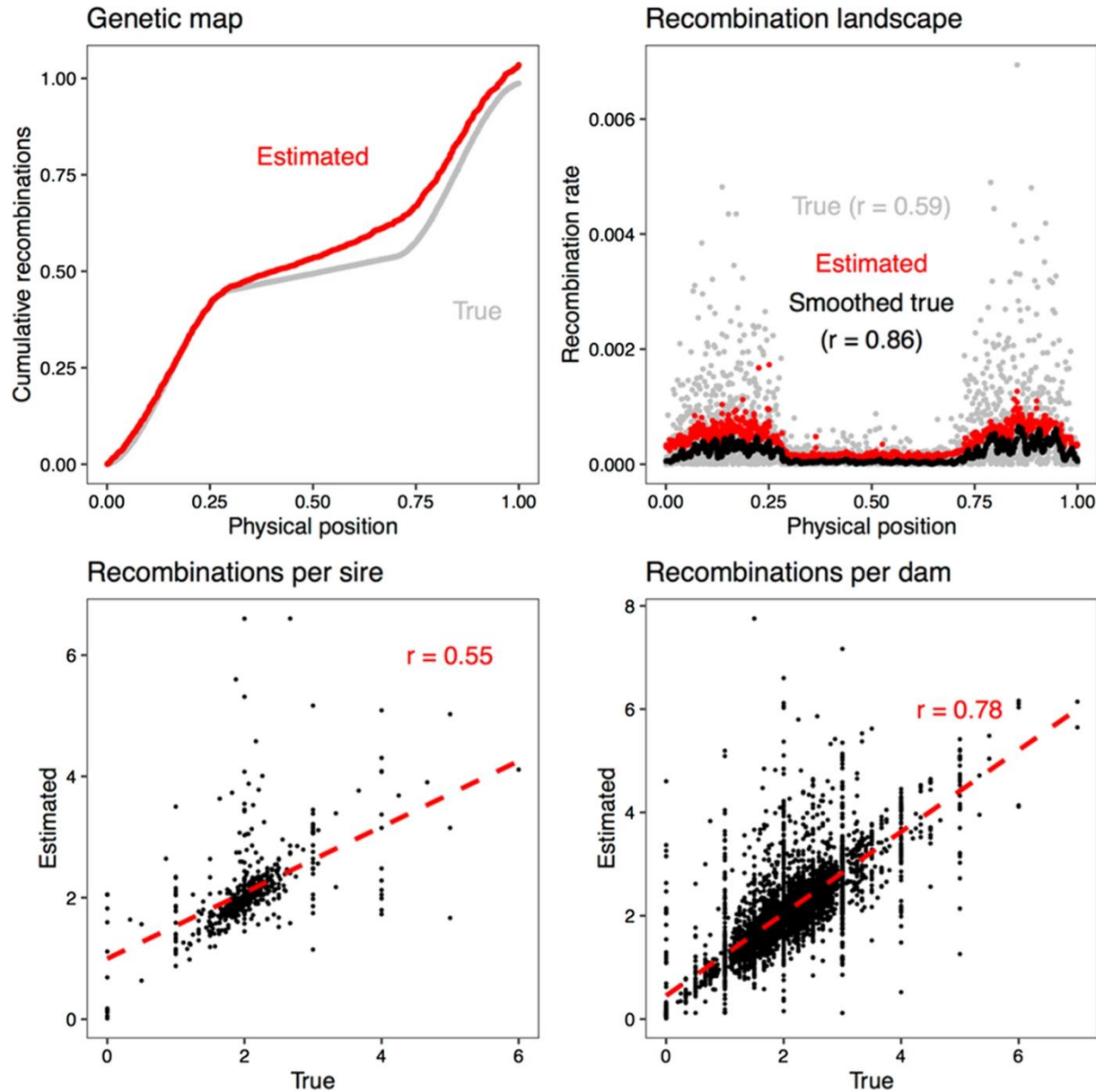


Integrating data and simulation to understand the effect of modern selective breeding

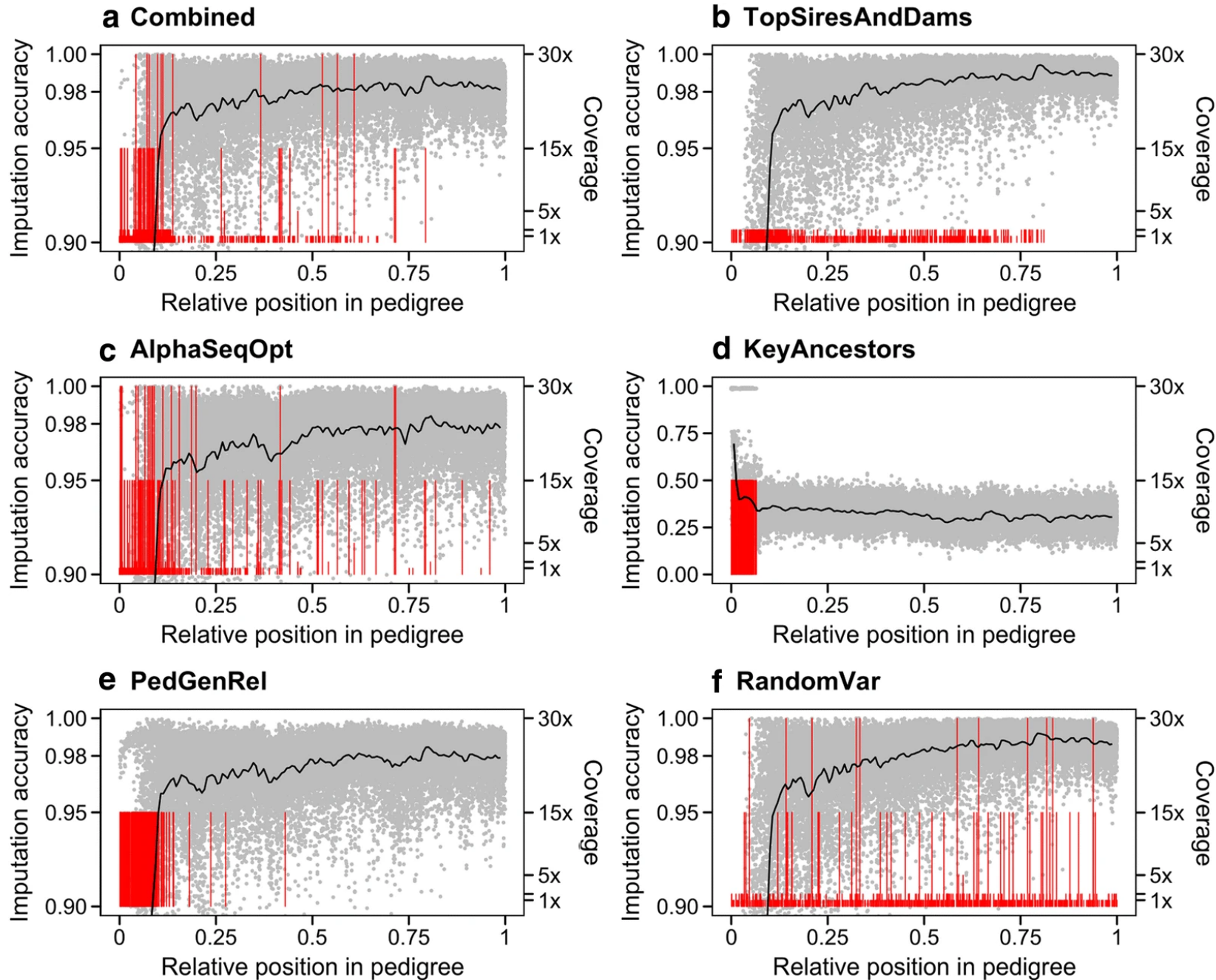
Empirical data, Wientjes et al. (2024)



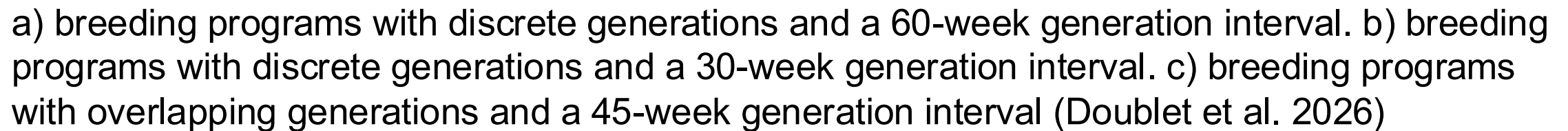
Allele frequency change of simulated causative variants, Wientjes et al. (2025)

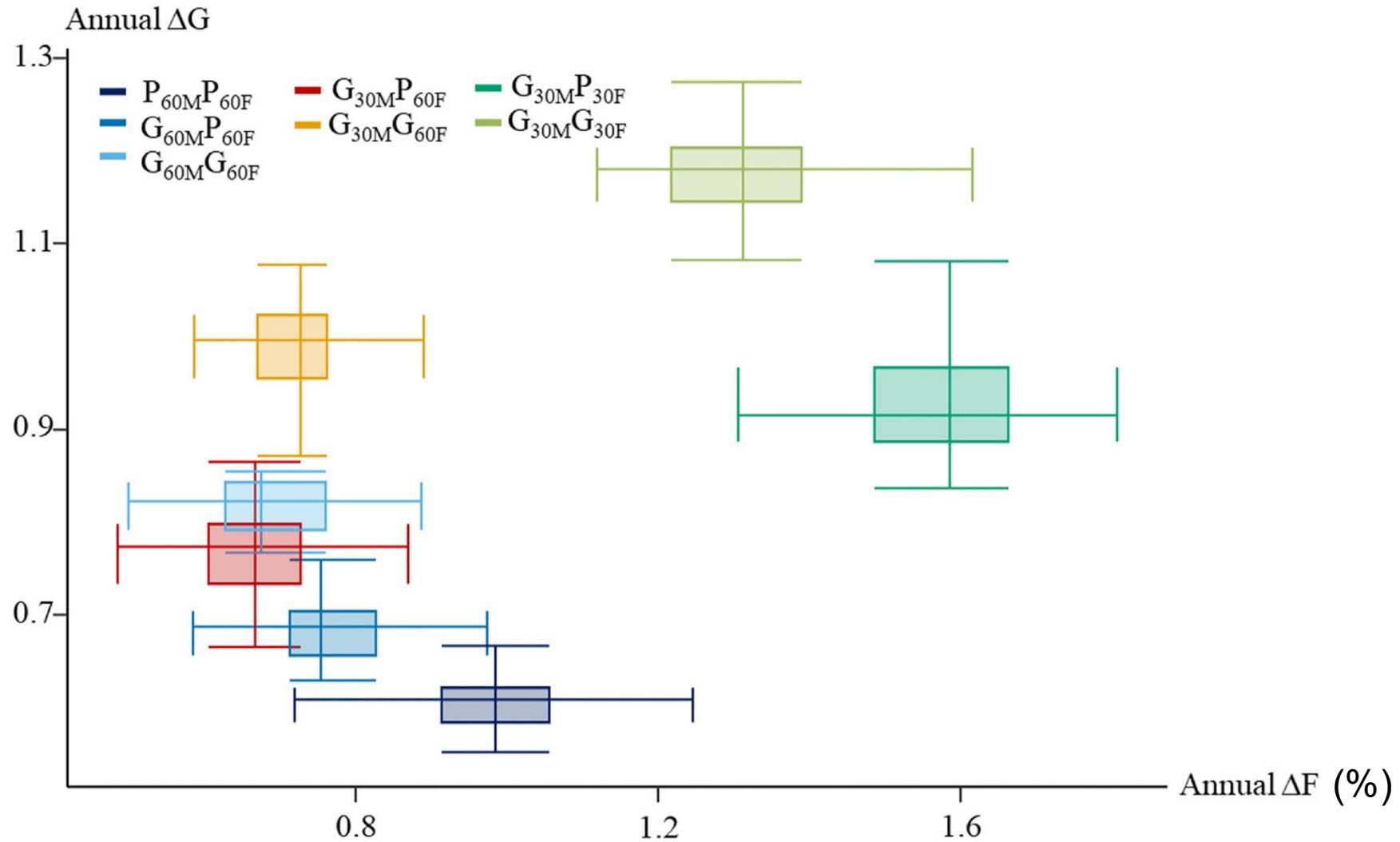


Evaluation of statistical methods, example from Johnsson et al. (2021; joint first with Amy Whalen)



Evaluation of sequencing strategies for imputation (Ros-Freixedes et al. 2020)





Evaluation of proposed breeding strategies on inbreeding rate and genetic gain (Doublet et al. 2026)

Applications of simulations

- Tackling research questions where we cannot gather data (yet or ever)
- Developing projects and designs
- Statistical model building for data analysis
- Evaluating statistical methods
- Simulation-based inference
- Self study, building our intuitions

Limitations of simulations

- Risk of getting stuck exploring many parameter combinations of little relevance to reality — “you can simulate anything; don’t simulate *everything*”
- Risk of simulating scenarios that are infeasibly expensive or complicated — “too much science fiction”
- Conclusions limited to cases that the model describes — “all models are wrong, but some of them are useful” (Box)

Approaches to genetic simulation

”Individual-based” simulation

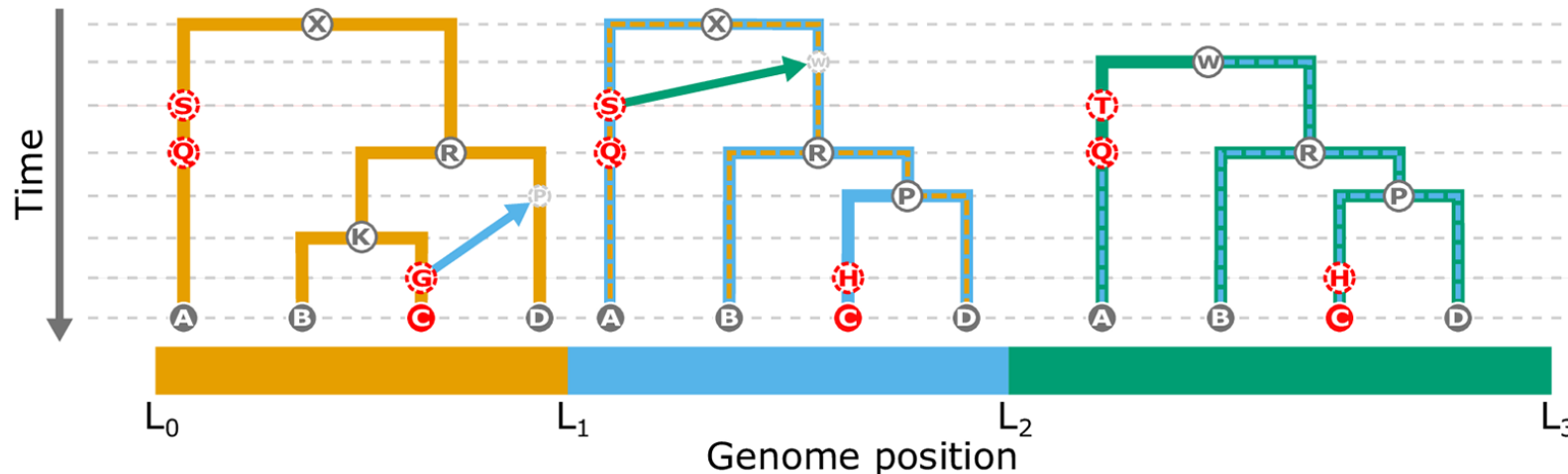
Most of our simulations are **individual-based**. They simulate animals and their genomes.

Simulations may be based on other objects: populations, equations for evolution of haplotypes, etc.

Forward and backward simulation

Most of our simulations run **forward in time**. One generation follows the other, like they do in real life.

Coalescent simulations run **backwards in time**, following haplotypes back in time until they “coalesce” into a common ancestor.



(Lewanski et al. 2024)

Population and quantitative genetic

Population genetic simulation: focus on genotypes, haplotypes, allele frequencies. Selection often absent or parametrised via fitness of genotypes.

Quantitative genetic simulation: focus on modelling complex traits with explicit phenotypes. Usually what we need for breeding work.

Tools for simulation

- Coalescent simulator: `msprime`
- Evolutionary forward simulator: `SLiM`
- Quantitative genetic forward simulator made for breeding: `AlphaSimR`
- Rolling your own (in R, Python etc)

Suggested reading: "Introduction to genetic simulation", course homepage, sections about `msprime`, `SLiM` and simple population genetic simulation in R.

There are many other good ones too!

Demos

A first AlphaSimR simulation

Goals:

- Demonstrate simulation structure and setup
- Demonstrate some features — what is possible
- Look at some R code

```
## coalescent simulate founder genomes
founders <- runMacs(nInd = 500,
                   nChr = 5)

## simulation settings
simparam <- SimParam$new(founders)
simparam$setSexes("yes_sys")
simparam$addTraitA(nQtlPerChr = 100)
simparam$setVarE(h2 = 0.4)

## generate the founder population
basepop <- newPop(founders,
                  simparam)
```

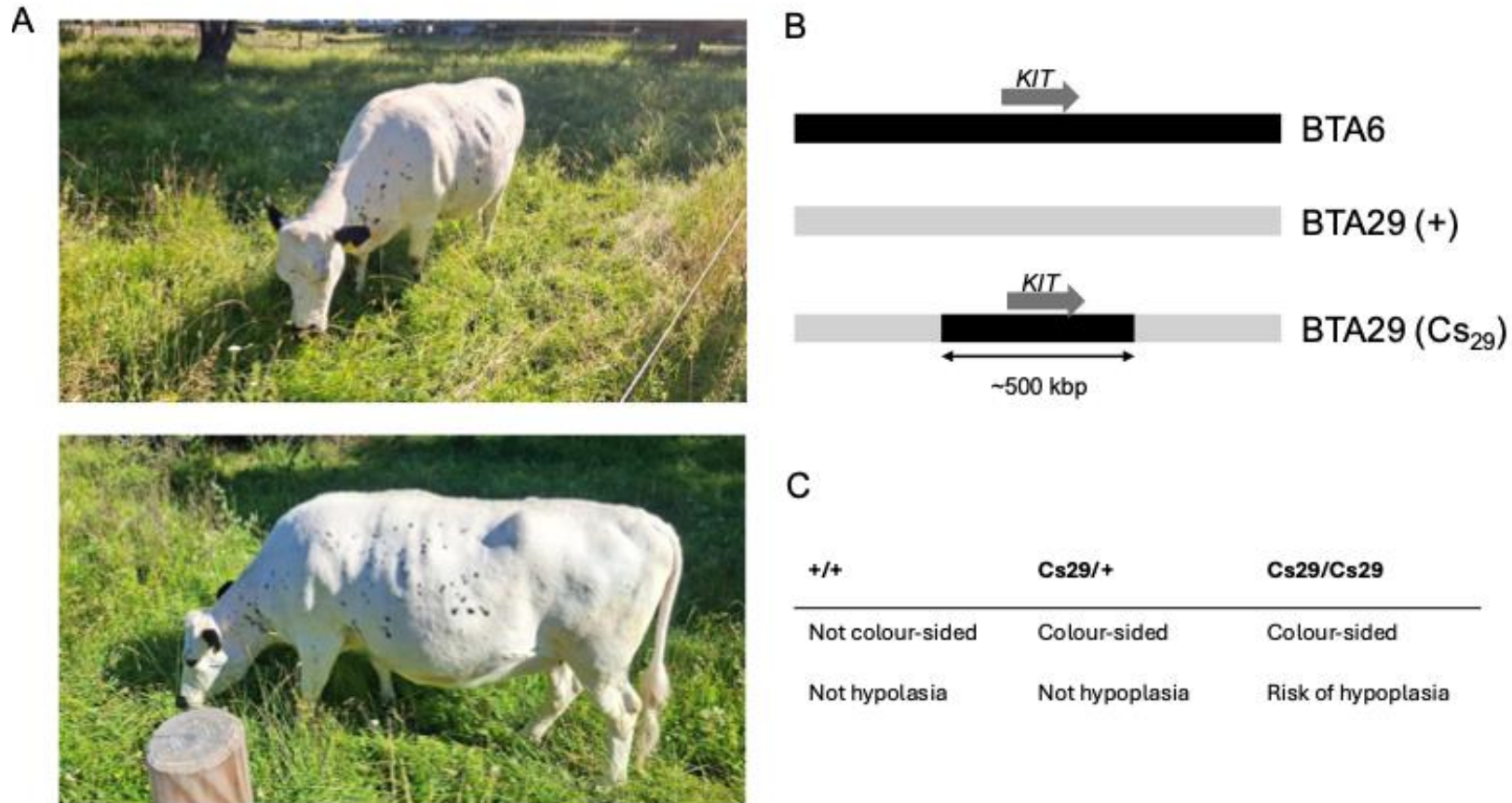
```
## data structures for the main breeding structure  
n_gen <- 40  
  
generations <- vector(length = n_gen,  
                        mode = "list")  
  
generations[[1]] <- basepop
```

```
## main breeding loop
for (gen in 2:n_gen) {
  sires <- selectInd(pop = generations[[gen - 1]],
                    nInd = 250 * prop_males,
                    simParam = simparam)
  dams <- selectInd(pop = generations[[gen - 1]],
                   nInd = 250,
                   simParam = simparam)
  offspring <- randCross2(females = dams,
                        males = sires,
                        nCrosses = 250,
                        nProgeny = 2,
                        simParam = simparam)
  generations[[gen]] <- offspring
}
```

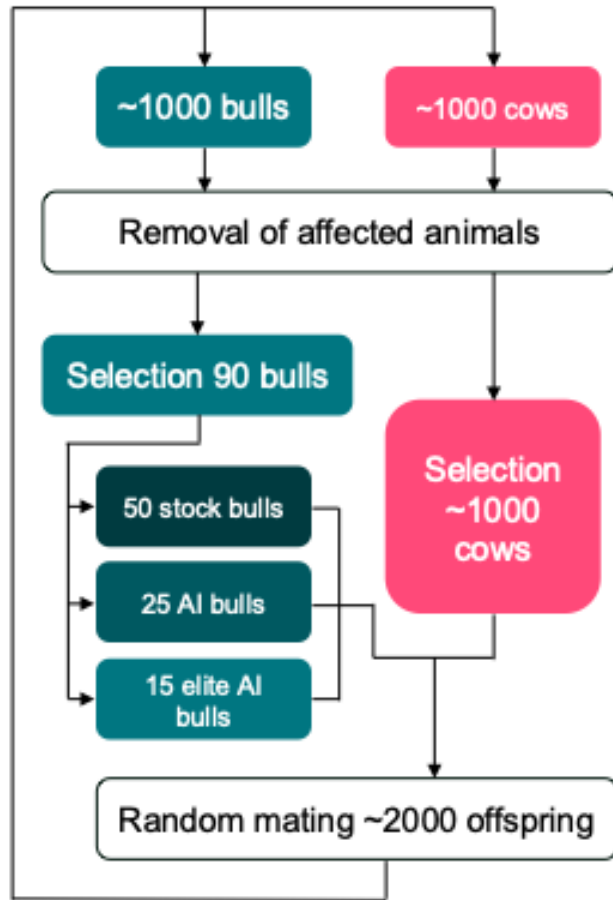
AlphaSimR features

- Extracting genotypes and haplotypes, pull*Geno, pull*Haplo
- Population object slots, @pheno, @gv, @ebv
- SimParam object parts, \$traits components
- Quantitative genetic parameters, meanG, varG, meanP, varP

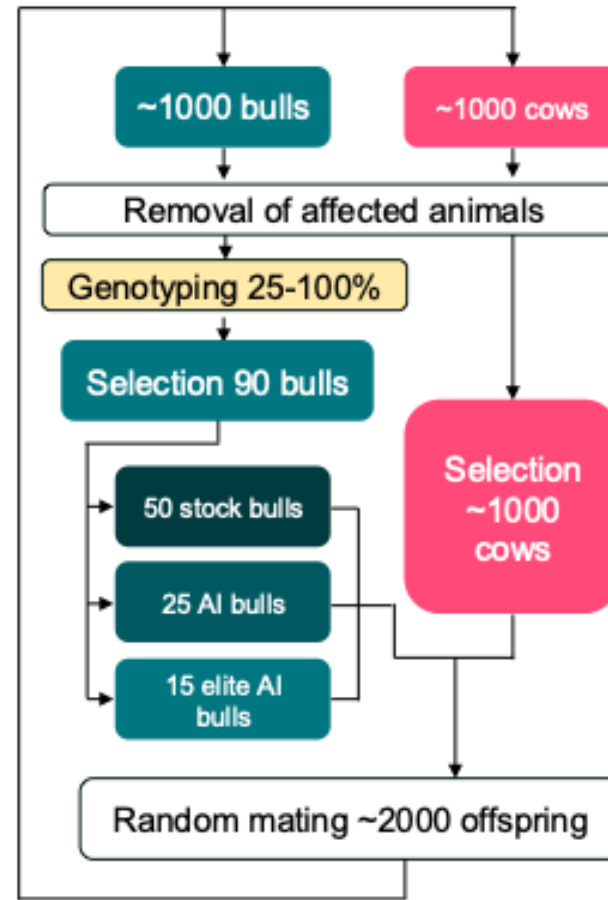
Example paper: Selection strategies against gonadal hypoplasia in Swedish Mountain cattle



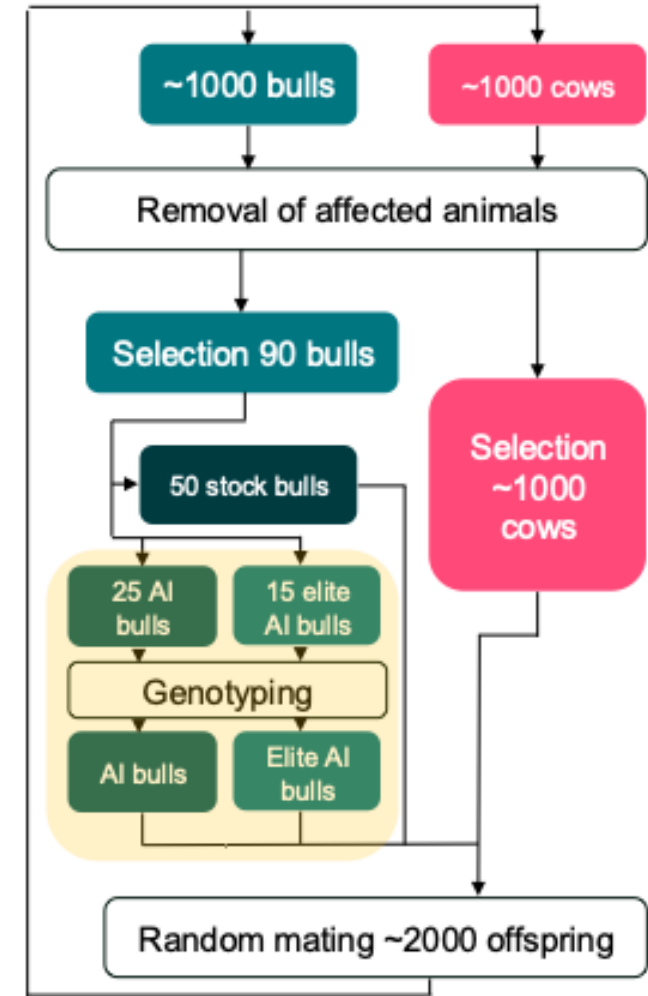
a)



b)



c)



Example paper: Gustafsson et al. (2026)

- Population genetic simulation in AlphaSimR
- Custom trait and selection code to simulate a **lethal allele with incomplete penetrance** and **marker-assisted selection**
- Recording allele frequencies, phenotype frequencies and inbreeding rate (to calculate inbreeding N_e)

Goals:

- Look at structure of a simulation project
- Get a feel for flexibility of AlphaSimR

Structure of a simulation in AlphaSimR

1. Define the founder population history.
2. Define simulation parameters.
3. Define genetic architecture.
4. Build the breeding structure.
5. Summarise results of the replicate.
6. Repeat for multiple replicates.
7. Collect results from multiple replicates.
8. Visualise outcomes.

Getting started with simulations

1. Clarify you are after. **What question are you trying to answer?** Why do you want to build a simulation?
2. Write preliminary specification (on paper, whiteboard, text file). **What do you want the simulation to do?**
3. Implement a part of the simulation. Start with a simple model and gradually add features.
4. Generate output from the preliminary model and inspect it. **Does the data (phenotypes, genotypes, genomic data) look sensible?**
Does the behaviour of the simulation agree with what you know?
5. Iterate from step 3, step 2, or even step 1 until you have added all the features that you need, and the results make sense to you.

Image credits

Margarita pizza <https://commons.wikimedia.org/wiki/File:Margwiki.jpg> Sarahinloes, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0/>>, via Wikimedia Commons. I have edited the image to take away the background, and this modified image is also licensed under <https://creativecommons.org/licenses/by-sa/4.0/>

Wientjes, Y.C.J., Peeters, K., Bijma, P. et al. Changes in allele frequencies and genetic architecture due to selection in two pig populations. Genet Sel Evol 56, 76 (2024). <https://doi.org/10.1186/s12711-024-00941-3> <http://creativecommons.org/licenses/by/4.0/>

Yvonne C J Wientjes, Piter Bijma, Joost van den Heuvel, Bas J Zwaan, Zulma G Vitezica, Mario P L Calus, The long-term effects of genomic selection: 2. Changes in allele frequencies of causal loci and new mutations, Genetics, Volume 225, Issue 1 (2025), <https://doi.org/10.1093/genetics/iyad141> <https://creativecommons.org/licenses/by/4.0/>

Ros-Freixedes, R., Whalen, A., Gorjanc, G. et al. Evaluation of sequencing strategies for whole-genome imputation with hybrid peeling. Genet Sel Evol 52, 18 (2020). <https://doi.org/10.1186/s12711-020-00537-7> <http://creativecommons.org/licenses/by/4.0/>

Johnsson, M., Whalen, A., Ros-Freixedes, R. et al. Genetic variation in recombination rate in the pig. Genet Sel Evol 53, 54 (2021). <https://doi.org/10.1186/s12711-021-00643-0> <http://creativecommons.org/licenses/by/4.0/>

Doublet M, Lecerf F, Pook T, Chapuis H, Raoul J, Herry F, et al. Stochastic simulations to optimize genomic selection for laying hens: Impact of generation interval and genotyping in the context of extended laying period. Poultry Science. 2026;105: 106870. <https://doi.org/10.1016/j.psj.2026.106870> <http://creativecommons.org/licenses/by/4.0/>

Lewanski AL, Grundler MC, Bradburd GS (2024) The era of the ARG: An introduction to ancestral recombination graphs and their significance in empirical evolutionary genomics. PLoS Genet 20(1): e1011110. <https://doi.org/10.1371/journal.pgen.1011110> <http://creativecommons.org/licenses/by/4.0/>



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